9.6 SlowFast Networks

- SlowFast networks can be described as a single stream architecture that operates at two different framerates, but the concept of pathways is used to reflect analogy with the biological Parvo- and Magnocellular counterparts
- the generic architecture has a Slow pathway and a Fast pathway, which are fused by lateral connections to a SlowFast network
- Figure 13 illustrates the concept

9.6 SlowFast Networks

- **Slow pathway**: the Slow pathway can be any convolutional model that works on a clip of video as a spatio-temporal volume
- the key concept in the Slow pathway is a large temporal stride τ on input frames, i.e., it processes only one out of τ frames
- a typical value of τ that was studied is 16—this refreshing speed is roughly 2 frames sampled per second for 30-fps videos
- denoting the number of frames sampled by the Slow pathway as T, the raw clip length is $T \times \tau$ frames

9.6 SlowFast Networks

- **Fast pathway**: in parallel to the Slow pathway, the Fast pathway is another convolutional model with the following properties
- **High frame rate**: the goal here is to have a fine representation along the temporal dimension
- the Fast pathway works with a small temporal stride of τ/α, where $\alpha > 1$ is the frame rate ratio between the Fast and Slow pathways
- the two pathways operate on the same raw clip, so the Fast pathway samples αT frames, α times denser than the Slow pathway
- a typical value is $\alpha = 8$ in the experiments
- the presence of α is in the key of the SlowFast concept (Figure 13, time axis)
- it explicitly indicates that the two pathways work on different temporal speeds, and thus drives the expertise of the two subnets instantiating the two pathways

9.6 SlowFast Networks

- **High temporal resolution features**: the Fast pathway not only has a high input resolution, but also pursues high-resolution features throughout the network hierarchy
- in the instantiations, no temporal downsampling layers are used (neither temporal pooling nor time-strided convolutions) throughout the Fast pathway, until the global pooling layer before classification
- as such, the feature tensors always have αT frames along the temporal dimension, maintaining temporal fidelity as much as possible

9.6 SlowFast Networks

- **Low channel capacity**: the Fast pathway also distinguishes with existing models in that it can use significantly lower channel capacity to achieve good accuracy for the SlowFast model; this makes it lightweight
- in a nutshell, the Fast pathway is a convolutional network analogous to the Slow pathway, but has a ratio of β ($\beta < 1$) channels of the Slow pathway
- the typical value is $\beta = 1/8$ in the experiments
- notice that the computation (floating-point operations, or FLOPs) of a common layer is often quadratic in terms of its channel scaling ratio
- this is what makes the Fast pathway more computation-effective than the Slow pathway
- in the instantiations, the Fast pathway typically takes $\sim$ 20% of the total computation
- interestingly, evidence suggests that $\sim$ 15-20% of the retinal cells in the primate visual system are M-cells (that are sensitive to fast motion, but not color or spatial detail)

9.6 SlowFast Networks

- the low channel capacity can also be interpreted as a weaker ability of representing spatial semantics
- technically, the Fast pathway has no special treatment on the spatial dimension, so its spatial modeling capacity should be lower than the Slow pathway because of fewer channels
- the good results of the model suggest that it is a desired tradeoff for the Fast pathway to weaken its spatial modeling ability, while strengthening its temporal modeling ability
- motivated by this interpretation, different ways of weakening spatial capacity in the Fast pathway are also explored, including reducing input spatial resolution and removing color information
- as is shown by experiments, these versions can all give good accuracy, suggesting that a lightweight Fast pathway with less spatial capacity can be made beneficial

9.6 SlowFast Networks

- **Lateral connections**: the information of the two pathways is fused, so one pathway is not unaware of the representation learned by the other pathway
- this is implemented by lateral connections, which have been used to fuse optical flow-based, two-stream networks
- in image object detection, lateral connections are a popular technique for merging different levels of spatial resolution and semantics
- one lateral connection is attached between the two pathways for every “stage" (Figure 13)
- specifically for ResNets, these connections are right after pool_1, res_2, res_3, and res_4

9.6 SlowFast Networks

- the two pathways have different temporal dimensions, so the lateral connections perform a transformation to match them
- unidirectional connections that fuse features of the Fast pathway into the Slow one (Figure 13) are used
- bidirectional fusion has been experimented with and similar results have been found
- finally, a global average pooling is performed on each pathway's output
- then, two pooled feature vectors are concatenated as the input to the fully-connected classifier layer

9.6 SlowFast Networks

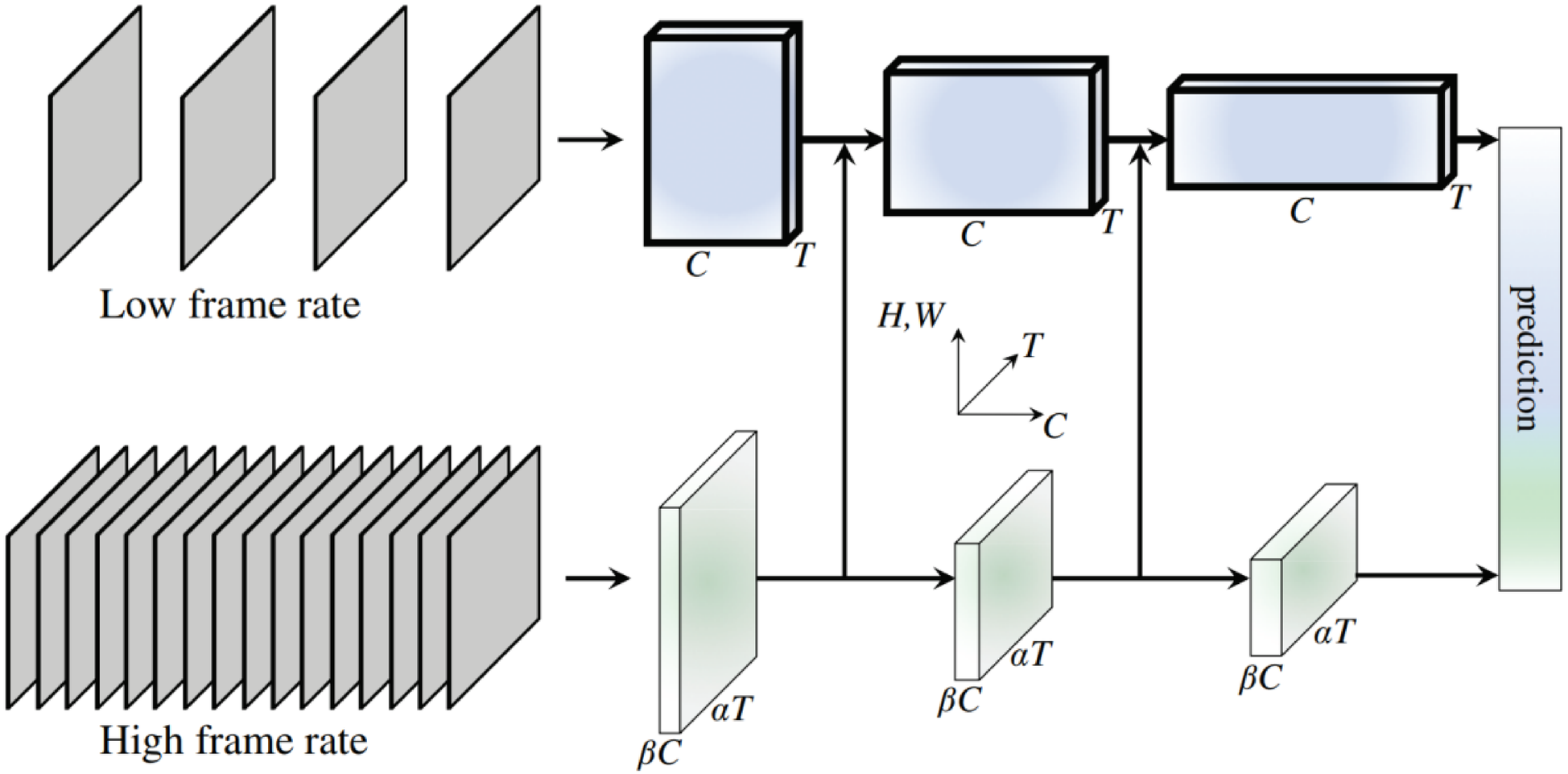


Figure 13: A SlowFast network has a low frame rate, low temporal resolution *Slow* pathway and a high frame rate, $\alpha\times$ higher temporal resolution *Fast* pathway. The Fast pathway is lightweight by using a fraction (β, e.g., $1/8$) of channels. Lateral connections fuse them.

9.6 SlowFast Networks

- the idea of SlowFast is generic, and it can be instantiated with different backbones and implementation specifics
- here, we describe the instantiations of the network architectures
- an example SlowFast model is specified in Figure 14
- we denote spatio-temporal size by $T \times S^2$, where T is the temporal length and S is the height and width of a square spatial crop
- the details are described next

9.6 SlowFast Networks

- **Slow pathway**: the Slow pathway in Figure 14 is a temporally strided 3D ResNet
- it has $T = 4$ frames as the network input, sparsely sampled from a 64-frame raw clip with a temporal stride $\tau = 16$
- temporal downsampling is not performed in this instantiation, as doing so would be detrimental when the input stride is large
- unlike typical C3D / I3D models, non-degenerate temporal convolutions (temporal kernel size > 1, underlined in Figure 14) are used only in res_4 and res_5; all filters from conv_1 to res_3 are essentially 2D convolution kernels in this pathway
- this is motivated by the experimental observation that using temporal convolutions in earlier layers degrades accuracy
- this is because, when objects move fast and the temporal stride is large, there is little correlation within a temporal receptive field unless the spatial receptive field is large enough (i.e., in later layers)

9.6 SlowFast Networks

- **Fast pathway**: Figure 14 shows an example of the Fast pathway with $\alpha = 8$ and $\beta = 1/8$
- it has a much higher temporal resolution (green) and lower channel capacity (orange)
- the Fast pathway has non-degenerate temporal convolutions in every block
- this is motivated by the observation that this pathway holds fine temporal resolution for the temporal convolutions to capture detailed motion
- further, the Fast pathway has no temporal downsampling layers by design

9.6 SlowFast Networks

- **Lateral connections**: the lateral connections fuse from the Fast to the Slow pathway
- it requires to match the sizes of features before fusing
- denoting the feature shape of the Slow pathway as $\{T, S^2, C\}$, the feature shape of the Fast pathway is $\{\alpha T, S^2, \beta C\}$
- the following transformations in the lateral connections are experimented with:
  1. (i) Time-to-channel: $\{\alpha T, S^2, \beta C\}$ is reshaped and transposed into $\{T, S^2, \alpha\beta C\}$, meaning that all α frames are packed into the channels of one frame.
  2. (ii) Time-strided sampling: one out of every α frames is sampled, so $\{\alpha T, S^2, \beta C\}$ becomes $\{T, S^2, \beta C\}$.
  3. (iii) Time-strided convolution: a 3D convolution of a 5×1^2 kernel with $2\beta C$ output channels and stride $= \alpha$ is performed.
- the output of the lateral connections is fused into the Slow pathway by summation or concatenation

9.6 SlowFast Networks

stage	*Slow* pathway	*Fast* pathway	output sizes $T \times S^2$
raw clip	-	-	64×224^2
data layer	stride 16, 1^2	stride **2**, 1^2	*Slow* : 4×224^2 *Fast* : 32×224^2
conv_1	1×7^2, 64 stride 1, 2^2	5×7^2, 8 stride 1, 2^2	*Slow* : 4×112^2 *Fast* : 32×112^2
pool_1	1×3^2 max stride 1, 2^2	1×3^2 max stride 1, 2^2	*Slow* : 4×56^2 *Fast* : 32×56^2
res_2	$\begin{bmatrix} 1 \times 1^2, 64 \\ 1 \times 3^2, 64 \\ 1 \times 1^2, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 3 \times 1^2, 8 \\ 1 \times 3^2, 8 \\ 1 \times 1^2, 32 \end{bmatrix} \times 3$	*Slow* : 4×56^2 *Fast* : 32×56^2
res_3	$\begin{bmatrix} 1 \times 1^2, 128 \\ 1 \times 3^2, 128 \\ 1 \times 1^2, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 3 \times 1^2, 16 \\ 1 \times 3^2, 16 \\ 1 \times 1^2, 64 \end{bmatrix} \times 4$	*Slow* : 4×28^2 *Fast* : 32×28^2
res_4	$\begin{bmatrix} 3 \times 1^2, 256 \\ 1 \times 3^2, 256 \\ 1 \times 1^2, 1024 \end{bmatrix} \times 6$	$\begin{bmatrix} 3 \times 1^2, 32 \\ 1 \times 3^2, 32 \\ 1 \times 1^2, 128 \end{bmatrix} \times 6$	*Slow* : 4×14^2 *Fast* : 32×14^2
res_5	$\begin{bmatrix} 3 \times 1^2, 512 \\ 1 \times 3^2, 512 \\ 1 \times 1^2, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} 3 \times 1^2, 64 \\ 1 \times 3^2, 64 \\ 1 \times 1^2, 256 \end{bmatrix} \times 3$	*Slow* : 4×7^2 *Fast* : 32×7^2
global average pool, concate, fc			# classes

Figure 14: An example instantiation of the SlowFast network. The dimensions of kernels are denoted by $\{T \times S^2, C\}$ for temporal, spatial, and channel sizes. Strides are denoted as {temporal stride, spatial stride^2}. Here the speed ratio is $\alpha = 8$ and the channel ratio is $\beta = 1/8$. τ is 16. The green colors mark *higher* temporal resolution, and orange colors mark *fewer* channels, for the Fast pathway. Non-degenerate temporal filters are underlined. Residual blocks are shown by brackets. The backbone is ResNet-50.